

# Kapruka Agent Challenge 2026 — Winning Design & Feature Spec

"Maatha / මාතෘ" — the trilingual diaspora gifting concierge that lets you *be present from 10,000 km away*.

|               |                                                                                      |
|---------------|--------------------------------------------------------------------------------------|
| Competition   | Kapruka Agent Challenge 2026 — "Build Sri Lanka's most innovative AI shopping agent" |
| Grand prize   | Apple M4 Mac mini                                                                    |
| Entries close | 30 June 2026                                                                         |
| Audience      | Sri Lankan developers                                                                |
| Doc date      | 8 June 2026                                                                          |
| Status        | Design locked — ready to scaffold Phase 1                                            |

## 1. Executive summary — the thesis that wins

Most entrants will build a chatbot that wraps the 7 Kapruka MCP tools in a chat loop. That is table stakes, not a winner.

The Kapruka MCP server quietly reveals what Kapruka actually *is*:

- It prices in **LKR, USD, GBP, AUD, CAD, EUR** — six currencies for a Sri Lankan store.
- Its order tracking exposes **delivery photo & video proof**.
- Its categories are full of **pirikara, dāna, Avurudu, Thai Pongal, Diwali, sympathies**.

That is not a generic store. **Kapruka's soul is the Sri Lankan diaspora sending gifts home** — a daughter in Melbourne sending her mother a birthday cake in Kandy, and never getting to see her face when it arrives.

So we don't build "a shopping agent." We build a **trilingual (Sinhala / Tamil / English + Singlish) diaspora gifting concierge with deep cultural intelligence** — festivals, almsgiving, gifting etiquette — and an **emotional payoff layer**: delivery proof, gift-message poetry, and occasion memory. That positioning is the differentiator.

## 2. Decision log (locked)

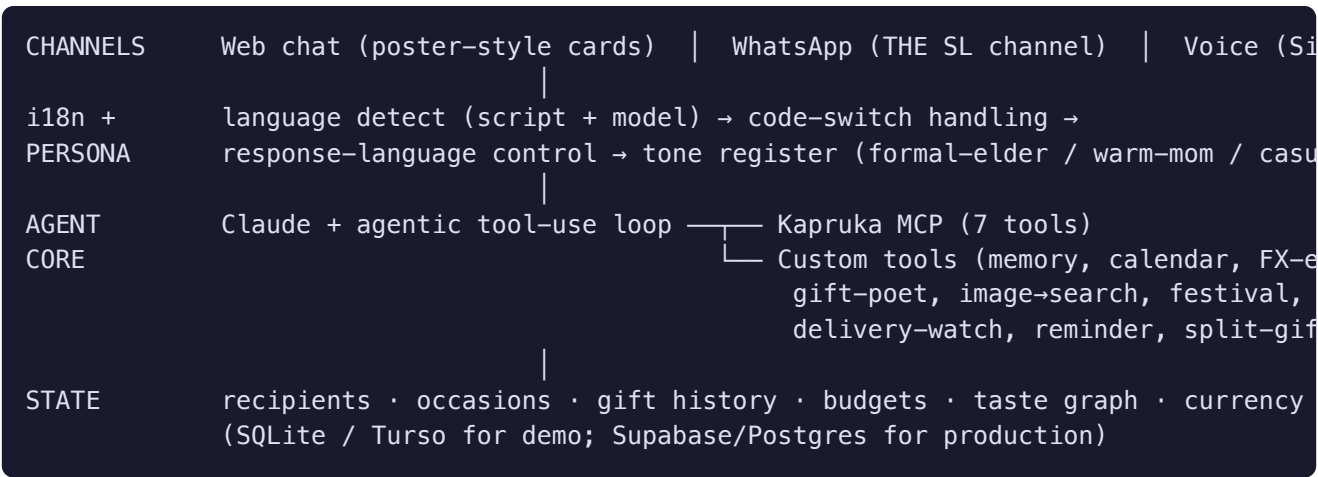
| Decision          | Choice                                                                                                                        | Why                                                                                                                                                                                        |
|-------------------|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AI provider / SDK | Claude (Anthropic Messages API) + manual agentic tool-use loop, Kapruka MCP connected. OpenAI Realtime added later for voice. | Best-in-class Sinhala/Tamil + cultural register; native MCP; prompt caching keeps the big cultural system prompt cheap; raw loop gives full control over streaming + custom product cards. |
| Stack / language  | TypeScript + Next.js (App Router).                                                                                            | Poster-style chat UI + server-side agent loop (API keys stay server-side) + easy deploy. Both SDKs and MCP are first-class in TS.                                                          |
| Channels          | Web chat + WhatsApp in parallel, over a shared agent core.                                                                    | Web mirrors the poster and demos cleanly; WhatsApp is THE Sri Lankan channel and the strongest "real product" signal.                                                                      |

## 3. The opportunity — why diaspora-first wins

- **The market is real and underserved.** Hundreds of thousands of Sri Lankans abroad (AU, UK, CA, US, EU) regularly send gifts home — birthdays, Avurudu, funerals, new babies. Kapruka is the dominant player precisely because of this.
- **The pain is emotional, not transactional.** They can't be there. They worry the cake will be stale. They don't know if it arrived. They want it to feel personal, in *their* language, in the right cultural register.
- **The MCP gives us exactly the primitives to solve it:** multi-currency, delivery-feasibility, perishability awareness, and delivery photo/video proof.

An agent that *understands this human story* — and speaks Sinhala the way Sri Lankans actually text — beats a faster product-search bot every time.

## 4. Architecture (4 layers)



## 5. Tech stack (concrete)

| Concern          | Choice                                                               |
|------------------|----------------------------------------------------------------------|
| Language         | TypeScript                                                           |
| App framework    | Next.js (App Router) — UI + server route handlers for the agent loop |
| Agent brain      | <code>@anthropic-ai/sdk</code> (Messages API), manual tool-use loop  |
| Tool surface     | Kapruka MCP server (MCP client) + custom in-process tools            |
| Frontend         | React + Tailwind + shadcn/ui; streaming chat with rich product cards |
| State / DB       | SQLite via Turso (demo) → Supabase/Postgres (prod)                   |
| WhatsApp channel | Twilio or WasenderAPI adapter over the shared agent core             |
| Voice (Phase 4)  | OpenAI Realtime API or Google Cloud STT/TTS ( <code>si-LK</code> )   |
| Deploy           | Vercel (web) + a small worker for reminders/delivery-watch           |
| Observability    | Structured logs, per-turn tool traces, cost/latency metrics          |

## 6. Kapruka MCP tool reference (the foundation)

All 7 tools verified live. Each supports `response_format: 'markdown' | 'json'` and currency `LKR/USD/GBP/AUD/CAD/EUR` .

| Tool                                      | Purpose                               | Key inputs                                                                                                                                                                                        | Notes / gotchas                                                                                                 |
|-------------------------------------------|---------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| <code>kapruka_list_categories</code>      | Top-level category tree + browse URLs | <code>depth</code> (1–2)                                                                                                                                                                          | ~65 categories incl. <code>pirikara</code> , <code>dāna</code> -adjacent, festival pages. Cached ~30 min.       |
| <code>kapruka_search_products</code>      | Keyword search + filters + pagination | <code>q</code> (min 3 chars), <code>category</code> , <code>min/max_price</code> , <code>in_stock_only</code> , <code>sort</code> , <code>cursor</code>                                           | <b>Pagination capped at 3 pages</b> (anti-enumeration). Stubs filtered unless <code>include_stubs=true</code> . |
| <code>kapruka_get_product</code>          | Full product detail                   | <code>product_id</code> , <code>currency</code> , <code>type</code>                                                                                                                               | Returns price, stock level, variants, images, shipping. Flags <code>CATSYM*</code> as non-purchasable.          |
| <code>kapruka_list_delivery_cities</code> | Find/validate deliverable cities      | <code>query</code> , <code>limit</code>                                                                                                                                                           | Returns canonical name + aliases (e.g. Kandy ← galagedara). Always pass a <code>query</code> .                  |
| <code>kapruka_check_delivery</code>       | Feasibility + flat rate for city/date | <code>city</code> , <code>delivery_date</code> , <code>product_id</code>                                                                                                                          | <b>Flat LKR rate per order.</b> Perishable warning fires for <code>CAKE*/FLOWER*/COMBO*</code> when >1 day out. |
| <code>kapruka_create_order</code>         | Guest checkout → pay link             | <code>cart[]</code> (1–30, qty 1–99, optional <code>icing_text</code> ), <code>recipient</code> , <code>delivery</code> , <code>sender</code> , <code>gift_message</code> , <code>currency</code> | <b>Only side-effecting tool.</b> Price locked 60 min; idempotency key per call. Rate limit 30 orders/hr/IP.     |
| <code>kapruka_track_order</code>          | Status + timeline + proof flags       | <code>order_number</code>                                                                                                                                                                         | Returns timeline, recipient, items, <code>has_delivery_photo/video</code> .                                     |

## Observed quirks (we handle these defensively — credibility points)

1. **`check_delivery` returns contradictory data.** In testing it reported *"Not available on this date"* while giving `next_available_date` = the *same* requested date. We never trust the availability boolean blind — we surface it honestly and re-confirm at order time.
2. **Product-ID casing is inconsistent** across tools ( `CAKE00KA001685` from search vs `cake00KA001685` from detail). Lookups are case-insensitive, but we **normalize IDs** before storing/deduping.
3. **`order_ref` ≠ `order_number`.** `create_order` returns a pre-payment `order_ref`; tracking needs the *post-payment* `order_number` emailed to the customer. We explain this gap to the user explicitly.

## 7. Custom tool surface (so it's clearly more than an MCP wrapper)

| Tool                                                             | Job                                                                        |
|------------------------------------------------------------------|----------------------------------------------------------------------------|
| <code>remember_recipient</code> / <code>get_recipients</code>    | Recipient address book (name, address, city, relationship, dietary, prefs) |
| <code>save_occasion</code> / <code>get_upcoming_occasions</code> | Per-user occasion calendar + reminder source                               |
| <code>record_gift_history</code> / <code>get_gift_history</code> | No-repeat intelligence (don't gift the same thing twice)                   |
| <code>schedule_reminder</code>                                   | Proactive nudges ("Amma's birthday in 5 days")                             |
| <code>convert_currency_explain</code>                            | Transparent FX with timestamp + source on top of MCP currency              |
| <code>compose_gift_message</code>                                | The Sinhala/Tamil/English gift-card "poet" (register-aware)                |
| <code>image_to_search_terms</code>                               | Multimodal: decompose an uploaded image into search queries                |
| <code>festival_calendar</code>                                   | SL festival + auspicious-time (nekath) knowledge                           |
| <code>gifting_etiquette</code>                                   | Relationship + religion-appropriate rules                                  |
| <code>delivery_proof_watch</code>                                | Polls <code>track_order</code> for photo/video, notifies the sender        |
| <code>split_gift_coordinate</code>                               | Shared-gift coordination over single-payer checkout (app-level)            |

## 8. THE FULL FEATURE CATALOG

### A. Core commerce (table stakes — must be flawless)

| #  | Feature                                               | Built on                          |
|----|-------------------------------------------------------|-----------------------------------|
| A1 | Natural-language product search (multilingual)        | <code>search_products</code>      |
| A2 | Category browse + deep-link to kapruka.com            | <code>list_categories</code>      |
| A3 | Product detail Q&A (price, stock, variants, shipping) | <code>get_product</code>          |
| A4 | Multi-item cart building (up to 30 items)             | local + <code>create_order</code> |
| A5 | Delivery feasibility (city + date + rate)             | <code>check_delivery</code>       |
| A6 | City resolver w/ alias + Singlish spelling tolerance  | <code>list_delivery_cities</code> |
| A7 | Guest-checkout pay-link generation w/ price breakdown | <code>create_order</code>         |
| A8 | Order tracking + timeline                             | <code>track_order</code>          |
| A9 | Live multi-currency (LKR/USD/GBP/AUD/CAD/EUR)         | all tools                         |

## B. Language & localization (the authenticity layer)

| #  | Feature                                                                                            |
|----|----------------------------------------------------------------------------------------------------|
| B1 | <b>Auto language detection</b> — Sinhala script, Tamil script, English, romanized "Singlish"       |
| B2 | <b>Code-switching</b> — understands "machan ammata mother's day ekata gift ekak ona" natively      |
| B3 | <b>Singlish transliteration</b> — "upandineta cake ekak" → "a birthday cake"                       |
| B4 | Respond in the user's language <i>and register</i> (formal Sinhala for elders, casual for friends) |
| B5 | Translate product names/descriptions into Sinhala/Tamil on request                                 |
| B6 | Sinhala date/number understanding ( "Avurudu welawata" , "dahas pahak" = 5000)                     |
| B7 | Sinhala/Tamil <b>voice in &amp; out</b> (accessibility for parents/grandparents)                   |

## C. Cultural intelligence (no competitor will go here)

| #  | Feature                                                                                                                                                                       |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| C1 | <b>Festival calendar engine</b> — Sinhala/Tamil New Year, Vesak, Poson, Esala, Deepavali, Thai Pongal, Christmas, Eid, Valentine's, Mother's/Father's Day                     |
| C2 | <b>Festival hamper auto-composer</b> — "Avurudu table" (kavum, kokis, kiribath fixings, sweetmeats) within budget, one delivery                                               |
| C3 | <b>Almsgiving / Dāna &amp; Pirikara concierge</b> — arrange pirikara to a temple or <i>mataka dāna</i> for a deceased relative (the MCP has a <code>pirikara</code> category) |
| C4 | <b>Gifting etiquette engine</b> — relationship + religion aware (no liquor to elders/monks, veg-appropriate, condolence vs celebration tone)                                  |
| C5 | <b>Auspicious-time (nekath)</b> awareness for New Year gifting                                                                                                                |
| C6 | <b>Sympathy / condolence mode</b> — tone-shifted flow + appropriate gifts ( <code>sympathies</code> category)                                                                 |

## D. Diaspora bridge (the emotional core)

| #  | Feature                                                                                                                  |
|----|--------------------------------------------------------------------------------------------------------------------------|
| D1 | <b>Timezone-aware</b> ("it's 2am in Colombo — I'll schedule the message for morning")                                    |
| D2 | <b>"Send to Amma"</b> recipient address book — names, addresses, relationships                                           |
| D3 | <b>Delivery photo/video proof loop</b> — surfaces <code>has_delivery_photo/video</code> so you see it arrive             |
| D4 | <b>Anonymous surprise mode</b> + heartfelt gift card ( <code>sender.anonymous</code> )                                   |
| D5 | <b>Recurring gifts</b> — "every Avurudu / every birthday, automatically"                                                 |
| D6 | <b>"Handle it all" delegated mode</b> — one sentence → agent picks, checks delivery, builds, pays-link                   |
| D7 | <b>Split-the-gift</b> — siblings abroad pool for one gift to parents (app-level coordination over single-payer checkout) |

## E. Proactive & agentic intelligence

| #  | Feature                                                                                                                   |
|----|---------------------------------------------------------------------------------------------------------------------------|
| E1 | <b>Occasion reminders</b> — "Amma's birthday in 5 days — repeat last year's cake, or something new?"                      |
| E2 | <b>No-repeat memory</b> — won't suggest the same gift two years running                                                   |
| E3 | <b>Flat-rate bundle optimizer</b> — exploits <i>one-flat-delivery-per-order</i> : "add flowers, delivery's still LKR 300" |
| E4 | <b>Budget optimizer</b> — best combo under "6000 incl. delivery"                                                          |
| E5 | <b>Perishable safety guard</b> — blocks scheduling a cake 5 days out without a freshness warning                          |
| E6 | <b>Delivery pre-check</b> — validates city/date <i>before</i> building the cart                                           |
| E7 | <b>Stock + price-lock urgency</b> — honest urgency from <code>stock_level</code> + the 60-min lock                        |

## F. Trust & safety (signals engineering maturity to judges)

| #  | Feature                                                                                                                       |
|----|-------------------------------------------------------------------------------------------------------------------------------|
| F1 | <b>Confirm-before-order</b> — <code>create_order</code> is the only side-effecting tool; always human-gated                   |
| F2 | <b>Full price transparency</b> — items + flat delivery + add-ons, every time                                                  |
| F3 | <b>Defensive delivery handling</b> — surface the <code>check_delivery</code> contradiction honestly, re-confirm at order time |
| F4 | <b>ID normalization</b> — handle <code>CAKE00KA...</code> vs <code>cake00KA...</code> casing across tools                     |
| F5 | <b>Idempotency + rate-limit awareness</b> (30 orders/hr, 30-min cache, no duplicate orders)                                   |
| F6 | <b>"Why this?"</b> — every recommendation explains itself                                                                     |

## G. Multimodal

| #  | Feature                                                                                            |
|----|----------------------------------------------------------------------------------------------------|
| G1 | <b>Image → search</b> — "find me something like this" from a photo                                 |
| G2 | <b>"Set the table from a photo"</b> — decompose a celebration image into purchasable Kapruka items |
| G3 | Gift-card / preview image generation                                                               |
| G4 | Voice-first mode for elders (Sinhala)                                                              |

## H. Memory & personalization

| #  | Feature                                                             |
|----|---------------------------------------------------------------------|
| H1 | Recipient profiles (relationship, dietary, preferences, past gifts) |
| H2 | Per-user occasion calendar                                          |
| H3 | Budget defaults & currency preference                               |
| H4 | Taste graph that learns from accepted/rejected picks                |

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## 9. The 6 headline innovations (the demo reel)

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1. **Trilingual code-switching incl. Singlish** — sounds like a real Sri Lankan, not Google Translate.
  2. **Diaspora "be there from afar"** — timezone + currency + delivery-photo proof + scheduled message.
  3. **Dāna / Pirikara almsgiving concierge** — nobody else will build this; profoundly Sri Lankan and emotionally huge for diaspora who miss funerals.
  4. **Cultural gift concierge** — festival + relationship + religion aware, with an Avurudu-hamper auto-composer.
  5. **Proactive occasion memory + no-repeat** — turns one purchase into a lifelong relationship.
  6. **Flat-rate bundle economics + honest data handling** — shows you understood the API better than the API understands itself.
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## 10. Sinhala / Tamil / multilingual deep-dive

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Sri Lanka is trilingual, and people **code-switch constantly** — including romanized Sinhala ("Singlish") over chat.

**Inputs the agent must handle natively:**

| Form                 | Example                                                     | Meaning                                             |
|----------------------|-------------------------------------------------------------|-----------------------------------------------------|
| Sinhala script       | අම්මාට උපන්දිනේට කේක් එකක් එවන්න ඕන                         | "Need to send a birthday cake for Amma"             |
| Singlish (romanized) | ammata bday ekata cake ekak Kandy yawanna puluwanda?        | "Can you send a birthday cake to Amma in Kandy?"    |
| Code-switch          | machan ammata mother's day ekata gift ekak ona, 6000ට යටින් | "Bro, need a Mother's Day gift for mum, under 6000" |
| Tamil                | அம்மாவுக்கு பிறந்தநாளுக்கு கேக் அனுப்ப வேண்டும்             | "Need to send a birthday cake for mum"              |

**Design rules:**

- **Detect** by script range first, fall back to the model for romanized/code-switched text.
  - **Mirror** the user's language *and register* — formal Sinhala for elders, warm for a parent, casual for a friend.
  - **Translate on demand** — product names/descriptions into Sinhala/Tamil when asked.
  - **Numbers & dates** — understand dahas pahak (5,000), Avurudu welawata (around New Year), relative dates.
  - **Voice (Phase 4)** — Sinhala STT is strong; TTS quality is the known weak link (we say so honestly).
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## 11. System prompt / persona design

The cultural brain lives in a cached system prompt. Core elements:

- **Identity:** warm, trustworthy Sri Lankan gifting concierge; speaks the user's language and register.
- **Mission:** help someone (often abroad) send the *right* gift home, and make them feel present.
- **Tooling discipline:** prefer `check_delivery` before building a cart; always show a full price breakdown; confirm before `create_order`.
- **Safety rules:** perishable guard; religion/relationship etiquette; never fabricate delivery dates; flag data quirks.
- **Cultural knowledge:** festival calendar, gifting customs, almsgiving conventions, auspicious timing.
- **Tone registers:** elder-formal / parent-warm / friend-casual / condolence-gentle.

## 12. Trust, safety & honest constraints

| Constraint                               | Mitigation                                                                                                   |
|------------------------------------------|--------------------------------------------------------------------------------------------------------------|
| No in-band payment confirmation          | Explain <code>order_ref</code> (pre-pay) vs <code>order_number</code> (post-pay, emailed); guide the handoff |
| <code>check_delivery</code> data flaky   | Defensive presentation; re-confirm at order time                                                             |
| Pagination capped at 3 pages             | Refine queries / filter by category instead of enumerating                                                   |
| Rate limits (30 orders/hr, 30-min cache) | Idempotency keys; cache-aware UX; never double-submit                                                        |
| Sinhala TTS limited                      | Text/STT strong; set expectations on voice output quality                                                    |
| Single-payer checkout                    | Split-gift is an app-level coordination layer, clearly labelled                                              |

## 13. Judge scoring map

| Likely criterion           | Features that score it                                                  |
|----------------------------|-------------------------------------------------------------------------|
| <b>Innovation</b>          | Dāna concierge, Singlish code-switch, diaspora bridge, bundle economics |
| <b>Cultural relevance</b>  | All of §C + §D, Sinhala/Tamil                                           |
| <b>Technical execution</b> | Agentic loop, custom tools, defensive data handling, MCP mastery        |
| <b>UX</b>                  | Poster-style cards, WhatsApp, voice, "why this"                         |
| <b>Completeness</b>        | Full purchase → track → proof → reminder loop                           |

## 14. The 90-second demo that wins

**User (Singlish):** machan amma ge bday ekata Kandy walata cake ekak yawanna ona, 60000 ශ්‍රී, surprise ekak

**Agent (Sinhala + English):**

1. Detects **Kandy** is deliverable ( `check_delivery` ).
2. Picks a top birthday cake ( `search_products` → `get_product` ).
3. **Flags the perishable timing** (freshness guard).
4. **Bundles flowers** because the flat delivery is the same LKR 300 (bundle optimizer).
5. Shows the total in **AUD** (diaspora currency).
6. **Writes a warm Sinhala gift-card message** ( `compose_gift_message` ).
7. Confirms, then returns the **pay link** ( `create_order` ).
8. Offers **delivery-photo confirmation** + "shall I remind you next Avurudu too?"

One flow, 8 differentiators, zero generic-chatbot energy.

## 15. Build roadmap (phased)

Locked decisions baked in: **TypeScript + Next.js, Claude brain + Kapruka MCP, Web + WhatsApp in parallel** over a shared agent core.

### Phase 1 — Demo-winning core

- Next.js app + shared agent core (Claude Messages loop + Kapruka MCP client).
- Poster-style web chat with streaming + product cards.
- Sinhala / Singlish detection + response.
- Cart → `check_delivery` pre-check → pay-link → track.
- Perishable guard, flat-rate bundle optimizer, "why this", full price breakdown.

### Phase 2 — Relationship layer

- Recipient/occasion memory + reminders + no-repeat.
- Festival calendar engine + Avurudu hamper composer.
- Dāna / pirikara flow.

### Phase 3 — Channels & proof

- WhatsApp channel adapter (Twilio / WasenderAPI).
- Delivery photo/video proof loop.
- Image → search (multimodal).

## Phase 4 — Voice & advanced

- Sinhala voice (OpenAI Realtime or Google `si-LK` ).
- Split-gift coordination.
- Taste graph personalization.

## Horizon 2+ — frontier vision

- See **§16 Frontier features** for the futuristic roadmap: agentic commerce, the recipient "emotional twin", Sinhala voice-note commerce, generative gifting, and group split-pay.

# 16. Frontier features — the futuristic horizon

Beyond the four build phases, these push the agent to the genuine 2026 frontier. Several are demo-ready; others are deliberately kept as **vision** so we signal ambition without over-promising to judges.

**Feasibility legend:**  demo-ready ·  ambitious but doable ·  research-grade / ethics-gated

## 16.1 Agentic commerce (the real 2026 frontier)

Aligned with the emerging agent-commerce standards (OpenAI's Agentic Commerce Protocol, Google's AP2 — Agent Payments Protocol). This is where the field is heading; judges who follow AI will recognise it instantly.

| Feature                               | What it is                                                                                                                                                       | Feasibility                                                                           |
|---------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| <b>Delegated autonomous gifting</b>   | Grant the agent a budget + policy (" $\leq$ LKR 8k per occasion, never liquor to Amma"); it executes whole occasions on its own, pinging only for final approval |  |
| <b>Standing gifting policies</b>      | "Flowers to Amma every month-end; a cake every birthday; pirikara every poya while I'm away" — the agent runs the calendar autonomously                          |  |
| <b>Agent-to-agent (A2A) checkout</b>  | Your agent transacts with Kapruka's agent under AP2-style payment mandates                                                                                       |  |
| <b>Computer-use checkout fallback</b> | The MCP only returns a <i>pay link</i> ; a browser-use sub-agent finishes payment under strict spend limits — closing the one gap in the MCP                     |  |

## 16.2 Predictive & anticipatory

| Feature                                   | What it is                                                                                                  | Feasibility |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------|-------------|
| <b>Life-event &amp; occasion radar</b>    | Predicts <i>who</i> you'll need to gift and <i>when</i> , before you ask (calendar, contacts, chat history) | 🟡           |
| <b>Relationship-health nudges</b>         | "You haven't sent your mother anything in 8 months" — gentle, opt-in                                        | 🟢           |
| <b>Pre-festival lock-ahead</b>            | "This cake sells out before every Avurudu — lock today's price now" (uses stock level + 60-min price lock)  | 🟢           |
| <b>Buy-now / deliver-later + FX hedge</b> | "AUD is strong this week — pay now, deliver on her birthday, price locked"                                  | 🟡           |

## 16.3 The emotional layer (the deepest moat)

| Feature                                      | What it is                                                                                                  | Feasibility                                                        |
|----------------------------------------------|-------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| <b>"Emotional twin" of each recipient</b>    | A learned taste model so you can ask " <i>what would Amma actually like?</i> " and get a real answer        | 🟡                                                                  |
| <b>Generative Sinhala card art + poetry</b>  | AI-made greeting-card image <i>with</i> a register-perfect Sinhala/Tamil verse, attached to the order       | 🟢                                                                  |
| <b>Personalized voice/video gift message</b> | A short AI-assisted video or voice note delivered with the gift                                             | 🔴 (voice-cloning = consent/ethics gate; keep assisted, not cloned) |
| <b>Grief companion mode</b>                  | For <i>mataka dāna</i> — a gentle, patient flow for diaspora who can't attend a parent's funeral almsgiving | 🟢                                                                  |
| <b>Reciprocity intelligence</b>              | "Aunty Kumari sent your kids gifts at Avurudu — reciprocate before Christmas?"                              | 🟡                                                                  |

## 16.4 Collaborative & social

| Feature                                   | What it is                                                                               | Feasibility |
|-------------------------------------------|------------------------------------------------------------------------------------------|-------------|
| <b>Live group-gifting split-pay</b>       | Siblings worldwide pool into one gift via a shared link; agent coordinates contributions | 🟡           |
| <b>Family anti-duplicate coordination</b> | The family sees what others are already sending Amma — no three cakes on one birthday    | 🟡           |
| <b>SL wedding/event registries</b>        | Agent-managed gift registry for diaspora weddings                                        | 🟡           |
| <b>Charity / dāna in someone's name</b>   | Donate to a Sri Lankan cause as the gift — culturally profound                           | 🟢           |
| <b>Community taste wisdom</b>             | Privacy-preserving "what others sent their mothers this Mother's Day"                    | 🔴           |

## 16.5 Spatial & generative experience

| Feature                               | What it is                                                                                            | Feasibility |
|---------------------------------------|-------------------------------------------------------------------------------------------------------|-------------|
| AI custom cake / arrangement designer | Describe it → generate the design image → custom Kapruka order.<br>The visual showstopper of the demo | 🟡           |
| AR preview in the recipient's room    | See the flowers on Amma's table before buying                                                         | 🔴           |
| Generative UI                         | The agent renders bespoke cards / comparisons per context instead of fixed templates                  | 🟢           |
| Virtual unwrapping                    | A delightful unwrap experience link for the recipient                                                 | 🟡           |

## 16.6 Voice-first & ambient (most Sri-Lanka-authentic)

| Feature                              | Why it matters                                                                  | Feasibility |
|--------------------------------------|---------------------------------------------------------------------------------|-------------|
| Sinhala WhatsApp voice-note commerce | Sri Lankans live on WhatsApp voice notes — "send Amma a cake," spoken not typed | 🟢           |
| Ambient agent                        | Lives in WhatsApp, learns passively, surfaces help proactively                  | 🟡           |
| Siri / Assistant shortcut            | "Hey Siri, send Amma flowers"                                                   | 🟡           |
| Recipient-side interaction           | Amma replies in Sinhala to confirm or choose a delivery time                    | 🟡           |

## 16.7 Reliability & logistics intelligence

| Feature                             | What it is                                                                                       | Feasibility |
|-------------------------------------|--------------------------------------------------------------------------------------------------|-------------|
| Auto-substitution on stockout       | Constraint-bounded ("swap to nearest cake ≤ budget; never change flavour family without asking") | 🟢           |
| Multi-recipient orchestration       | "Send to all 5 cousins for Avurudu" in one pass                                                  | 🟢           |
| Weather / monsoon-aware timing      | Adjust delivery date around forecast disruption                                                  | 🟡           |
| Live tracking + proactive ETA pings | Real-time status to the sender abroad                                                            | 🟡           |

## 16.8 Trust, safety & provenance

| Feature                            | What it is                                             | Feasibility |
|------------------------------------|--------------------------------------------------------|-------------|
| Verified delivery                  | Recipient OTP / photo confirm before "delivered"       | 🟢           |
| Diaspora scam / fraud detection    | Flags suspicious requests and unusual patterns         | 🟡           |
| Explainability "receipt"           | Every autonomous action logged with <i>why</i>         | 🟢           |
| Privacy-preserving recipient vault | Encrypted address book; PII never leaves where it must | 🟢           |

## 16.9 The on-device meta-angle 🎁

| Feature                 | Why it's clever                                                                                                                                                                                                                                             |
|-------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| On-device privacy layer | Run the sensitive memory / PII reasoning locally via a small model; cloud only for heavy lifting.<br>Pitch: <i>"the privacy-critical brain runs entirely on-device — on the very M4 Mac mini you're giving away."</i> A narrative the judges will remember. |

## 16.10 The five that win — and the combined demo

If we build only five frontier features, build these:

1. **Delegated autonomous gifting + standing policies** (§16.1) — proves real agency, not chat.
2. **"Emotional twin" — "what would Amma like?"** (§16.3) — nobody else will have it.
3. **Sinhala WhatsApp voice-note commerce** (§16.6) — most authentically Sri Lankan, most accessible.
4. **AI custom cake designer → order** (§16.5) — the visual showstopper.
5. **Live group-gifting split-pay** (§16.4) — solves a real, unsolved diaspora pain.

**They stack into one killer demo:** *speaks* a request in Sinhala on WhatsApp → agent consults the emotional twin → designs a custom cake → coordinates a sibling split-pay → executes autonomously under your budget policy → delivers photo proof.

## 16.11 Ethics & scope guardrails

- **Voice / video cloning, social-media enrichment, AR, and community taste wisdom** are kept as **vision slides**, not shipped features — they carry consent, privacy, or maturity risks. Showing them as roadmap signals ambition without over-promising.
- Anything autonomous (§16.1) ships **only** with explicit spend limits, an approval gate, and a full explainability receipt (§16.8).

## 17. Data model (initial)

|               |                                                                                                       |
|---------------|-------------------------------------------------------------------------------------------------------|
| recipients    | id, user_id, name, relationship, phone, address, city, dietary, preferences[], religion, notes        |
| occasions     | id, user_id, recipient_id, type, date, recurrence, currency                                           |
| gift_history  | id, user_id, recipient_id, occasion_id, product_id, product_name, ordered_at, order_ref, order_number |
| reminders     | id, user_id, occasion_id, fire_at, status                                                             |
| user_prefs    | user_id, default_currency, language, timezone                                                         |
| taste_signals | id, user_id, recipient_id, product_id, signal (accepted/rejected)                                     |

## 18. Appendix — cultural data seeds

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**Festival calendar (seed):** Sinhala & Tamil New Year (Apr), Vesak (May), Poson (Jun), Esala Perahera (Jul/Aug), Deepavali (Oct/Nov), Thai Pongal (Jan), Christmas (Dec), Eid (lunar), Valentine's (Feb), Mother's/Father's Day.

**Gifting etiquette (seed rules):**

- No alcohol to monks, elders by default, or religious occasions.
- Condolence/sympathy: muted tone, no celebratory items, white flowers appropriate.
- New baby: practical + sweet; avoid sharp objects (superstition).
- Boss/corporate: neutral, premium, non-personal.

**Almsgiving (dāna) conventions:** pirikara packs for temple offerings; *mataka dāna* (almsgiving in memory of the deceased) — sensitive register, often arranged by diaspora unable to attend.

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*Prepared for the Kapruka Agent Challenge 2026. Built on the live Kapruka MCP server (7 tools verified). Ready to scaffold Phase 1 on approval.*